The Abdus Salam International Centre for Theoretical Physics

IAEA International Atomic Energy Agency

unesco United Nations Educational, Scientific and Cultural Organization

Workshop on Impurity Problems in Quantum Mixtures: from Ultracold Gases to Electron Matter

Description:

How the properties of impurities are modified by their interaction with a surrounding medium is a fundamental question across several research areas and different physical systems. Yet, a common set of ideas and technical tools can be applied to these seemingly different frameworks, the investigation of one system yielding information on another.

MORE DETAILS:

This conference aims to bring together leading experts in the multi-disciplinary study of impurity phenomena in a broad range of quantum systems, from ultracold atomic gases to strongly correlated electronic materials and heterostructures.

TOPICS:

- Impurities and Polarons in Ultracold Quantum Gases
- Excitons, Polarons, and Exciton-Polaritons in 2D and Solid-State Systems
- Polaron Formation, Dynamics, and Non-Equilibrium Phenomena
- Impurities with Long-Range or Tunable Interactions
- Impurities as Quantum Probes of Many-Body Systems
- Quantum Simulation and Information Approaches to Impurity Physics

The workshop schedule will combine pedagogical and cross-disciplinary presentations with more technical ones, leaving ample time for spontaneous discussions and thus fostering new collaborations and approaches in the study of impurity problems in emerging quantum science platforms.

SPEAKERS:

J. Arlt
Aarhus University, Denmark
C. Baroni
IQOQI Innsbruck, Austria
M. Berciu
University of British Columbia, Canada
A. Bergschneider
University of Bonn, Germany
G. Bruun
Aarhus University, Denmark
C. Chin
University of Chicago, USA
X. Cui
Chinese Academy of Sciences, China
E. Demler
ETH Zurich, Switzerland
C. Eigen
University of Cambridge, UK
D. Fausti
UNITS, Italy and FAU Erlangen, Germany
R. Fletcher
MIT, USA
C. Franchini
University of Vienna, Austria and University of Bologna, Italy
D. Golež
University of Ljubljana, Slovenia
A. Imamoglu
ETH Zurich, Switzerland
M. Lebrat
University of Harvard and JILA, USA
P. Massignan
Polytechnic University of Catalonia, Spain
A. Recati
CNR-INO, Italy
R. Schmidt
Heidelberg University, Germany
S. Skipetrov
LPMMC and CNRS, France
J. Tempere
University of Antwerp, Belgium
A. Widera
RPTU Kaiserslautern-Landau and State Research Center OPTIMAS, Germany

23 - 25 February 2026

Trieste, Italy

Application and Deadlines:

For applicants requesting financial and/or visa support:
30 November 2025

For all other applicants:
7 January 2026

DIRECTORS:

L. Ardila
University of Trieste, Italy
F. Scazza
University of Trieste, Italy
M. Zaccanti
Istituto Nazionale di Ottica del Consiglio Nazionale delle Ricerche (CNR-INO) and European Laboratory for Non-Linear Spectroscopy (LENS), Italy

LOCAL ORGANISER:

M. Dalmonte, ICTP, Italy

FURTHER INFORMATION:

E-mail: smr4208@ictp.it

Web: http://indico.ictp.it/event/11134/

Female scientists are encouraged to apply.

GRANTS:

A limited number of grants are available to support the attendance of selected participants, with priority given to participants from developing countries.
There is no registration fee.



LENS European Laboratory for Non-Linear Spectroscopy